SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct: (192424426L)

I S.V.DiLiP *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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HR

CO 1 - AT 2

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1) Backtracking Search Employee task Assignment.

* Problem understanding

A college must assign 4 employee (E_1, E_2, E_3, E_4) to 4 administrative tasks.

* Admissions (A)
* Accounts. (AC)
* Record (R)
* IT Support (IT)

Each employee must be assigned exactly one task and each task must have only one employee.

Constraints.

1) E_1 ~~an~~ can only hande Admission or Accounts.

2) E_2 must be assigned before E_3 according to task order.

3) only one employee per task.

4) E4 cannot handle Records.

5) Every employee must receive exactly one task.

Constraint Analysis.

Constraint	Analysis.
E1 → Admission or Account.	Limits E1 choice to two task
E2 before E3	Position of E2 must come earlier than E3 in task order
One employee per task	Prevents duplicate. assignments.
E4 ≠ Records.	Records cannot be assigned to E4
Every employee. exactly one task	complete assignment. required.

Solution Development Backtracking Steps

Step 1.

Assign

E1 -> Admission

Remaning task:

Accounts, Records, IT

Step 2

Try

E2 -> IT support.

Remaining:

Accounts, Records.

Check constraint.

task order.

Admission -> Accounts. -> Records - IT

Step 3

Try

E2 -> Accounts.

Remaining..

Records IT.

constraint satisfied.

4) Application of concepts

The Backtracking Search Algorithm Systematically explores possible assignments.

Whenever a constraint is violated, the algrithm immediately backtrack and tries another assignment.

This reduces unnecessary search and guarntees a valid solution.

5) Justification.

The final assignment stisfis all constrin.

*) E1 assigned only to Admission.

*) E2 assigned before E3

*) One employee per task

*) E4 is not assigned records.

*) Every employee assigned exactly one task.

BFS Search on a 5×5 Grid.

Problem understanding.

A delivery robot must travel from Start (S) to Goal (G) in a 5×5 city grid.

Movement is allowed in four directions.

* Up
* Down.
* Lift.
* Right.

Each movement costs 1 unit.

Some road are one-way.

Block direction cannot be crossed.

Assumed Grid.

S	.	.	.	.
.	x	.	x	.
.	.	.	.	.
x	.	x	.	.
.	.	.	.	G

S - Start.

G - Goal.

x - Block cell.

Constraint Anlysis &

Constraint	Analysis.
up, Down, left	Robot can move only in four.
cost = 1	Uniform cost per movement.
oneway streets	Restrited direction. cannot be used.
manhttam heuristic	Estimates distance using row and column.

4) Application of concepts.

The Breath-First Search (BFS) algorithm escpands nodes level by level,

Since very move has equal cost

(1) BFS. garntee the shorts path.

The manhattan distance help estimate the remaining distance to the gal.

Justification.

*) Explores nodes level-wise

*) never revisits visited nodes.

*) Respects movement restriction.

*) Find the shortest path

*) total path cost = 8, which is optimal